



# Koneru Lakshmaiah Education Foundation

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## DEPARTMENT OF CSE

### PROJECT ABSTRACT SUBMISSION

### COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

| TITLE OF THE PROJECT |               |                 |
|----------------------|---------------|-----------------|
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## Abstract

The **AI-Powered Autonomous Bug Detection and Code Quality Analyzer** is an intelligent software engineering system designed to automatically identify bugs, code quality issues, security vulnerabilities, and potential performance problems in source code. The system uses Artificial Intelligence and Machine Learning techniques to analyze code, detect patterns associated with defects, and provide meaningful recommendations for improving software quality.

The proposed system supports the principles of Adaptive Software Engineering by continuously analyzing changing code and adapting its recommendations based on newly detected defects and coding patterns. Developers can upload or submit source code, after which the analyzer performs automated static analysis and AI-based inspection to identify syntax errors, logical issues, code smells, duplicated code, security risks, and maintainability problems. The system generates a quality report containing the detected issues, their severity, possible causes, and recommended fixes.

An AI-based feedback mechanism can prioritize critical issues and suggest suitable code improvements, reducing the time and effort required for manual code review. The system can also track code quality across different versions, allowing developers to observe improvements or degradation over time. By integrating automated bug detection, intelligent recommendations, and continuous quality monitoring, the proposed system aims to improve software reliability, maintainability, and development efficiency while supporting adaptive and iterative software development practices.

Signature of the Guide

HOD